

Module 7: Lecture 21

Some Additional Analysis

. **Keywords.** Capturability analysis

8.3 Some Additional Analysis

8.3.1 The nonlinear engagement block in the BR/CLOS guidance loop

The block in the guidance loop shown in Figures 8.5 and 8.8 are obtained after making several strong assumptions. We will see this below:

From Figures 8.4 or 8.7, we have

$$R_M \dot{\theta}_M = V_M \sin(\alpha_M - \theta_M) \quad (8.32)$$

Differentiating,

$$\dot{R}_M \dot{\theta}_M + R_M \ddot{\theta}_M = V_M \cos(\alpha_M - \theta_M)(\dot{\alpha}_M - \dot{\theta}_M) \quad (8.33)$$

Since, $\dot{R}_M = V_M \cos(\alpha_M - \theta_M)$ and $\dot{\alpha}_M = a_M/V_M$,

$$2V_M \cos(\alpha_M - \theta_M) \dot{\theta}_M + R_M \ddot{\theta}_M = a_M \cos(\alpha_M - \theta_M) \quad (8.34)$$

From which, we can write,

$$a_M = 2V_M \dot{\theta}_M + \frac{R_M}{\cos(\alpha_M - \theta_M)} \ddot{\theta}_M \quad (8.35)$$

Since $V_M \cos(\alpha_M - \theta_M)$ is the component of the missile velocity along the LOS, we can make an assumption that

$$\frac{R_M}{V_M \cos(\alpha_M - \theta_M)} = \tilde{t} \quad (8.36)$$

where, $\tilde{t}$ maybe an approximation of the time elapsed (which is t). This is in the same spirit as that used to compute an approximate value of time-to-go. Then,

$$a_M = 2V_M \dot{\theta}_M + V_M \tilde{t} \ddot{\theta}_M = V_M \tilde{t} \left(\frac{2}{\tilde{t}} \dot{\theta}_M + \ddot{\theta}_M \right) \quad (8.37)$$

If $\tilde{t}$ is the time elapsed then $V_M \tilde{t} = \tilde{R}_M$ is the distance covered by the missile. Thus,

$$a_M = \tilde{R}_M \left(\frac{2}{\tilde{t}} \dot{\theta}_M + \ddot{\theta}_M \right) \quad (8.38)$$

Taking Laplace transforms on both sides,

$$\frac{\theta_M(s)}{a_M(s)} = \frac{1}{\tilde{R}_M s \left(\frac{2}{\tilde{t}} + s \right)} \quad (8.39)$$

Now, we make a further strong assumption that the time elapsed is large enough for us to neglect the term $2/\tilde{t}$ and also that $\tilde{R}_M \cong R_M$. (Note that these are very strong assumptions). Then,

$$\theta_M(s) = \frac{1}{R_M} \cdot \frac{1}{s^2} a_M(s) \quad (8.40)$$

This is the relationship used to linearize the non-linear block. Hence, while evaluating the guidance law in the loop, we should use the nonlinear engagement geometry rather than the linear representation.

8.3.2 Capturability analysis

We can carry out some limited capturability analysis here. For this, we need to first determine the conditions at the initial time and final time. From (8.35), we get an expression for the missile latax,

$$a_M = 2V_M \dot{\theta} + \frac{R_M}{\cos(\alpha_M - \theta)} \ddot{\theta} \quad (8.41)$$

At $t = 0$, we have $R_M = 0$, and so,

$$a_{M0} = 2V_M \dot{\theta}_0 \quad (8.42)$$

Now, for perfect LOS guidance to be effective, (8.5) has to hold at initial time too. This means that

$$V_M \sin(\alpha_{M0} - \theta_0) = 0 \Rightarrow \alpha_{M0} = \theta_0 \quad (8.43)$$

which implies that initially the missile must point towards the target.

Similarly, suppose at some time $t = t_f$ interception takes place, then $R_f = 0$, and from (8.9),

$$R_{Mf} = \sqrt{R_0^2 + (V_T t_f)^2 + 2R_0 V_t t_f \cos(\theta_0 - \alpha_T)} \quad (8.44)$$

From (8.5),

$$V_M \sin(\alpha_{Mf} - \theta_f) = V_T \sin(\alpha_{Tf} - \theta_f) \quad (8.45)$$

which, from (8.4) implies that

$$V_{\theta Rf} = 0 \quad (8.46)$$

which is an obvious condition for interception.

Theorem 8.1. If $V_M > V_T$ then the missile captures the target.

Proof. We will prove this by contradiction. Assume that interception does not occur. In that case the target continues to fly for infinite time. Now, as $t \rightarrow \infty$, we have $\theta \rightarrow \alpha_T$ which, from (8.5), implies that $\alpha_{M\infty} = \theta_\infty = \alpha_T$. This implies that the missile converges to a tail-chase situation with the target as $t \rightarrow \infty$. From (8.2) and (8.4), we have $V_{\theta\infty} = 0$ and $V_{R\infty} = V_T - V_M$. Now, if $V_M > V_T$, then the missile will eventually intercept the target. This is a contradiction. And so, the missile will intercept the target in finite time whenever $V_M > V_T$. $\square$

This result can be interpreted as follows: The theorem gives a sufficient condition only. Based on the sufficient condition one can say that the capture region for the LOS guidance is at least as large as that of the pure pursuit guidance law, provided that the initial conditions are the ideal initial conditions for LOS guidance, that is the initial missile velocity vector points directly at the target (note that these initial conditions are the same as that of pure pursuit). However, the theorem does not give a necessary condition and so we cannot say if $V_M \leq V_T$ implies no interception. There could be conditions under which M may intercept T even when $V_M \leq V_T$.

8.4 Concluding Remarks

In this chapter we analyzed another classical guidance law called the LOS guidance law. We also discussed the implementation of the LOS guidance law in two forms: Beam Rider and Command to LOS. Their relative performances were also discussed. However, note that unlike in the previous chapters we did not carry out any detailed

capturability analysis in the relative velocity space. This is because, the LOS guidance equations do not lend themselves very easily to an analysis in the relative velocity space. Moreover, capturability of LOS guidance laws depends on other parameters than just the relative velocity components. In fact, complete capturability results can be obtained only through numerical computations.

Questions

Assume initial position of approaching target: (8 to 12 kms down range, 0.5 to 0.8 km altitude)

1. Consider a BR missile

- Assume that there is no autopilot compensation:

(A) Guidance law is $a_M = K R_M(\theta_T - \theta_M)$. Consider $K = 1, 5, 10$. $V_T/V_M = 0.6$. Plot the following quantities against time (a) missile latax (b) missile and target trajectories (c) $R_M(\theta_T - \theta_M)$ (d) R = Separation between missile and target. Find the miss distance in each case.

(B) Let $K = 10$ and $V_T/V_M = 0.6$. Assume a pitch up constant target maneuver with $a_T = 3g$. Plot against time (a) missile latax (b) R (c) missile and target trajectories. Find the miss-distance.

- Assume that there is autopilot compensation:

$$G(s) = K \left(\frac{1 + s/2}{1 + s/20} \right)$$

(C) Assume a non-maneuvering target. Let $K = 10$, $V_T/V_M = 0.6$. Plot against time (a) missile latax (b) R (c) missile and target trajectories. Find the miss-distance.

(D) Assume a similar maneuvering target. Let $K = 10$, $V_T/V_M = 0.6$. Plot against time (a) missile latax (b) R (c) missile and target trajectories. Find the miss-distance.

2. Consider a CLOS missile:

(A) Without autopilot compensation and $K = 1, 5, 10$, $V_T/V_M = 0.6$. The guidance law is given by $a_M = K R_M(\theta_T - \theta_M) + R_M \ddot{\theta}_T + 2\dot{R}_M \dot{\theta}_T$.

For a non-maneuvering target and then for a maneuvering target (3g pitch-up maneuver):

Plot against time (a) missile latus (b) missile target trajectories (c) R . Find the miss-distance in each case.

(B) Do the same as above for a case with autopilot compensation $G(s)$ given in the BR case.

Write overall comments comparing the results.

References

1. C.-F. Lin: *Modern Navigation, Guidance, and Control Processing*, Prentice Hall, Englewood Cliffs, NJ, 1991.
2. A.S. Locke: *Guidance*, D. Van Nostrand Co., 1955.
3. P. Zarchan: *Tactical and Strategic Missile Guidance*, 3rd Ed., AIAA Inc., Washington DC, 1997.